

## SYSTEM FOUNDRY / PERSONAL MVP

# FridgeFirst

A local-first use-first queue that makes forgotten food and its purchase value visible.

|                          |                 |                  |
|--------------------------|-----------------|------------------|
| PRODUCT STATUS           | AUTOMATED TESTS | ONE-TIME PRICE   |
| <b>PASS</b>              | <b>8 / 8</b>    | <b>KRW 8.9K</b>  |
| Implemented and verified | Zero failures   | Personal license |

## 01 / PROJECT OVERVIEW

## Project overview

FridgeFirst turns a small household refrigerator into a use-first queue. Users register food with an expiry date and purchase value, see what should be handled within three days, and record whether each item was consumed or wasted.

## 02 / TARGET CUSTOMER

## Target customer

Solo residents and one-to-two-person households that buy groceries in small batches, forget partially used food, and want a simpler tool than meal planning or full pantry software.

## 03 / PROBLEM

## Problem

Food becomes invisible behind newer purchases. A plain expiry list does not show the money at risk, while memory and paper labels do not preserve whether waste is actually improving.

PRODUCT

# Three focused jobs

04 / CORE FEATURES

## Core features

The MVP implements one complete household food-control loop:

|    |                                                                                                                                         |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Validated food inventory<br>Capture name, category, quantity, unit, purchase value, and expiry date with strict limits.                 |
| 02 | Use-first queue and value at risk<br>Order expired, today, soon, and later items; total purchase value for items due within three days. |
| 03 | Consumed and wasted history<br>Remove handled items and separately total value used versus discarded.                                   |

05 / HOW TO USE

## How to use

After grocery shopping, register each food item. Open the queue before planning a meal, handle orange items first, then mark each item consumed or wasted. The header shows both current risk and the value successfully used.

06 / SYSTEM ARCHITECTURE

## System architecture

A static browser UI calls pure validation and queue functions, then persists inventory and history in localStorage. No account, backend, database, analytics, or external API is used.

| Browser UI | Inventory module | Local storage    | Node tests     |
|------------|------------------|------------------|----------------|
| HTML + CSS | Rules + sorting  | Device-only data | Pure functions |

07 / TECHNOLOGY

## Technology

Static HTML, responsive CSS, vanilla JavaScript, localStorage, and the built-in Node.js test runner. Runtime dependencies: zero. Paid APIs: zero.

## VALIDATED INTERFACE

# Actual running screen

Captured from headless Chromium after a strawberry item was validated and added to the queue.

The screenshot displays the FridgeFirst application interface. At the top, the logo 'FF FridgeFirst EAT WHAT MATTERS FIRST' is visible on the left, and a '샘플 복원' button is on the right. A message states: '식품을 등록하고 소비 우선순위를 경선했습니다.' Below this, the section 'PERSONAL FOOD CONTROL' features a large heading '냉장고에서 먼저 쓸 것만 보여줘요.' and a sub-message: '소비기한이 가까운 식품과 위험 금액을 한눈에 보고, 사용·폐기를 기록해 낭비를 줄이세요.' A circular badge on the right says 'LOCAL FIRST 가가 판매만 지원'.

Four summary cards are shown: '등록 식품 6 ITEMS IN FRIDGE', '3일 내 우선 소비 4 USE FIRST', '낭비 위험 금액 ₩21,700 VALUE AT RISK', and '사용 완료 금액 ₩0 처리 기록이 없습니다'. Below these is the 'USE-FIRST QUEUE 소비 우선순위' table, which lists items with their expiration dates, names, purchase prices, and status buttons ('사용 완료' and '폐기 기록').

On the right, the 'ADD FOOD 식품 등록' section includes a note: '영수증을 버리기 전에 20초만 기록하세요.' It has input fields for '식품명' (set to '딸기'), '분류' (set to '채소·과일'), '소비기한' (set to '08/07/2026'), '수량' (set to '1'), '단위' (set to '팩'), and '구매가' (set to '5000'). A '우선순위 계산·등록' button is at the bottom. A small note at the bottom right says: '식품 안전 안내 표시 기한만으로 안전을 보장하지 않습니다. 상태가 의심되면 섭취하지 마세요.'

| 종류     | 소비기한          | 상품명    | 구매가    | 상태    |
|--------|---------------|--------|--------|-------|
| 오늘까지   | 2026-08-04 1  | 샐러드 채소 | ₩6,500 | 사용 완료 |
| 1일 남음  | 2026-08-05 2  | 그릭 요거트 | ₩4,500 | 사용 완료 |
| 2일 남음  | 2026-08-06 1  | 부침용 두부 | ₩2,800 | 사용 완료 |
| 3일 남음  | 2026-08-07 2팩 | 딸기     | ₩7,900 | 사용 완료 |
| 7일 남음  | 2026-08-11 10 | 달걀     | ₩7,200 | 사용 완료 |
| 28일 남음 | 2026-09-01 1  | 냉동 만두  | ₩9,800 | 사용 완료 |

## 08 / RUNNING SCREEN

## Running screen

The verified screen shows six items, four due within three days, and KRW 21,700 at risk after adding two packs of strawberries worth KRW 7,900. Marking that item consumed returned the active inventory to five items and increased used value to KRW 7,900.

## VERIFICATION

# Evidence, not assumptions

## 09 / TEST RESULTS

## Test results

All required checks were executed against the generated project.

| Check                   | Result | Evidence                                           |
|-------------------------|--------|----------------------------------------------------|
| Dependency installation | PASS   | npm install completed; zero runtime dependencies   |
| Application start       | PASS   | Local HTTP server returned status 200              |
| Core workflow           | PASS   | Risk KRW 13.8K to 21.7K; used value to KRW 7.9K    |
| Automated tests         | 8 / 8  | Validation, date ordering, totals, outcome history |
| README commands         | PASS   | Install, test, and start commands executed         |
| Credential scan         | PASS   | No secret patterns and no .env file                |
| PDF                     | PASS   | Generated, parsed, rendered, visually inspected    |

## 10 / SECURITY AND LIMITATIONS

## Security and limitations

FridgeFirst sends no data outside the browser. Text, numeric, date, category, and outcome inputs are validated and displayed text is HTML-escaped. It requires no credentials. Device-local data is unencrypted and disappears when browser storage is cleared. Expiry priority is an organization aid, not a food-safety guarantee; users should not consume food whose condition is uncertain or whose safe-use guidance says otherwise.

## GO TO MARKET

# A small habit with visible value

## 11 / PRICING AND REVENUE MODEL

## Pricing and revenue model

Sell a downloadable personal license for KRW 8,900 as a one-time purchase. The price stays below the value of one typical forgotten grocery item and avoids hosting costs. The MVP contains no paid service or subscription dependency.

## 12 / FIRST CUSTOMER ACQUISITION

## First customer acquisition

Recruit the first twenty users from Korean solo-living, meal-prep, and zero-waste communities. Offer a 14-day test and ask users to photograph their next grocery receipt, then compare recorded at-risk and discarded value. Convert with the concrete promise of seeing which purchase needs attention first.

## 13 / IMPROVEMENT PLAN

## Improvement plan

Priorities after observing real refrigerator routines:

- JSON backup and restore for device migration.
- Barcode-assisted item entry using an optional local catalog.
- Recurring staples and partial-quantity updates.
- Private encrypted sync only if users request multiple-device access.

## PRODUCT BOUNDARY

FridgeFirst should remain a low-friction inventory decision tool. It should not make food-safety diagnoses, prescribe diets, or become a full recipe marketplace.

ASSESSMENT

# Overall evaluation

14 / OVERALL SCORE

## Overall score

Six dimensions are scored out of 10. A higher maintenance score means easier maintenance.

| Dimension                 | Score | Rationale                                                                    |
|---------------------------|-------|------------------------------------------------------------------------------|
| Customer problem strength | 8/10  | Forgotten groceries create recurring, visible household waste.               |
| One-day completeness      | 9/10  | The full register, prioritize, and outcome loop is browser-tested.           |
| Sales potential           | 7/10  | A low one-time price fits the audience, but retention must be observed.      |
| Maintenance difficulty    | 9/10  | Zero runtime dependencies and no backend keep maintenance light.             |
| Differentiation           | 8/10  | Purchase-value risk is clearer than a generic expiry list.                   |
| Scalability               | 8/10  | Barcode entry, quantities, backups, and optional sync offer expansion paths. |

OVERALL

8.2 / 10

Proceed to a twenty-user trial and measure discarded value.

Verdict: technically complete and visually polished for a personal MVP. The next useful evidence is whether users keep registering groceries after the first week and reduce recorded waste.